Vinay Babannavar

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Professional Summary

Computer Science professional with hands-on experience in full-stack web development and AI/ML systems. Proficient in building end-to-end web applications using HTML, CSS, JavaScript, React.js, REST APIs, Node.js, MongoDB, and MySQL. Experienced in backend development with Python (FastAPI, Flask, Django), version control, and CI/CD pipelines. Strong problem-solving skills with a track record of building and deploying intelligent, production-ready applications. Looking to contribute to innovative full-stack and AI-driven products.

Education

T. John Institute of Technology — Bengaluru, India
Bachelor of Engineering in Computer Science, CGPA: 9/10.0 — *Aug 2023 – Jun 2027*

Govindram Seksaria Science College — Belagavi, India
Pre-University Education (Science), 80% — *Jun 2021 – May 2023*

Technical Skills

- **Web Development:** HTML, CSS, JavaScript, React.js, Node.js, REST APIs, Responsive UI Design
- **Full Stack Development:** React.js (Frontend), Python – Django/Flask/FastAPI (Backend), REST API Integration
- **Programming Languages:** Python, Java
- **Backend Frameworks:** Django, Flask, FastAPI
- **Databases:** MongoDB, MySQL, PostgreSQL
- **Tools & Platforms:** Git, GitHub, VS Code, Docker, Jenkins, CI/CD
- **AI / Machine Learning:** Machine Learning, LLM APIs, RAG, Agentic AI
- **Soft Skills:** Design Thinking, Problem Solving, Team Collaboration, Public Speaking

Experience

Web Development Intern — Unified Mentor — *Apr 2026 – Jul 2026*

– Completed a three-month web development internship, contributing to the design and development of production-oriented web applications.
– Developed **SwapWear**, a MERN-stack web application, as the primary project, implementing responsive UI, frontend functionality, and backend integration.
– Applied **React.js, Node.js, Express.js, MongoDB, REST APIs, and Git** throughout the development lifecycle.

Publications

Towards a Connected Campus: AI-Augmented Institutional Management Platform *May 2026*
IJSRED, ISSN: 2581-7175, Vol. 9, Issue 3 – Co-authored and published a peer-reviewed paper.

Projects

RecruitMind AI — Intelligent Candidate Discovery *2026*
Python, LangGraph, ChromaDB, FastAPI, Streamlit

– Engineered a stateful 6-node LangGraph orchestration pipeline to parse job descriptions, compute 384-dimensional dense embeddings, and query 300+ candidate profiles via ChromaDB.

AI-Powered Jenkins Pipeline Analyzer & Self-Healing DevOps Agent *2026*
Python, Jenkins, LLM APIs, CI/CD

– Developed an AI agent that analyzes Jenkins CI/CD pipeline logs and automatically identifies build failures, errors, and misconfigurations.

MediLedger — GST-Compliant Pharmacy Billing & Inventory System *2026*

React, FastAPI, PostgreSQL, Tailwind CSS, Docker Compose

– Built a full-stack, GST-compliant billing and inventory management platform for a retail pharmacy using React 19 and FastAPI, with PostgreSQL for persistent transactional data.

Self-Reflective Corrective RAG Pipeline *2026*

Python, LangGraph, DeepSeek-R1, ChromaDB, Pinecone, FastAPI

– Engineered a corrective RAG system using LangGraph with a stateful graph of 5 nodes – retrieval, relevance grading, web search fallback, generation, and hallucination checking – enabling full self-correction without human intervention.

Intelligent Digital Twin System for Smart Technologies *2025*

Python, IoT Sensors, Machine Learning

– Developed a digital twin simulation system that syncs real-time IoT sensor data with machine learning models for continuous monitoring.

Smart AI Traffic Signal System *2025*

Python, OpenCV, Machine Learning

– Developed an AI-driven traffic signal system using computer vision to detect vehicle density and optimize signal timing dynamically.

Certifications

– Microsoft: Azure AI Fundamentals (AI-900), Azure Fundamentals (AZ-900)
– TATA Forage: GenAI Powered Data Analytics Job Simulation
– Cisco: Networking Basics
– NPTEL: Cloud Computing, Database Management System, Introduction to Machine Learning
– Outskill: Generative AI Mastermind
– Others: Innovation & Design Thinking Program, Programming in C